

LATCHANA DEVI M V

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B.Tech, Electronics and Communication Engineering · Batch 2027 · Thiruvapur, Tamil Nadu

PROFESSIONAL SUMMARY

Final year student with strong foundation in real-time embedded control, system design, and electronics. Proficient in microcontroller programming (STM32, Arduino, ESP32) with demonstrated ability to design, implement, test, and optimize complete control systems. Strong understanding of control algorithms, real-time systems, multi-sensor integration, and system optimization. Proven systematic debugging and problem-solving approach. Excellent communication skills through published research and technical presentations. Demonstrated technical leadership and collaboration. Seeking Control Systems Engineering Intern role to apply embedded control expertise and grow specialized systems engineering capabilities.

CONTROL SYSTEMS & ELECTRICAL ENGINEERING SKILLS

Control Systems: System optimization, Performance tuning, Embedded control, Feedback systems, Multi-sensor control

Electronics & Embedded Systems: STM32, Arduino, ESP32 microcontrollers, Circuit design, Multi-sensor integration, Sensor calibration, Real-time systems, Hardware-software integration

Electrical Engineering: Signal processing, Analog & digital circuits, System architecture, Electrical integration

Professional Competencies: Problem-solving, Systematic debugging, Root cause analysis, Communication, Collaboration, Technical documentation, Attention to detail, Learning agility

CONTROL SYSTEMS ENGINEERING PROJECTS

EV Charging Station Management System — Real-Time Control Systems Design & Optimization

STM32 microcontroller, Real-time control logic, Multi-sensor integration, Feedback control, Performance optimization, System calibration

- Designed complete control system architecture combining multi-sensor data acquisition (voltage/current sensors) with real-time feedback control logic and relay switching management
- Implemented real-time control algorithms with sensor calibration, threshold-based logic, and adaptive parameter tuning; optimized system performance achieving ~15% improvement
- Applied systematic debugging and control system troubleshooting; identified sensor calibration offset → implemented firmware corrections → validated performance through rigorous testing

Air Quality Monitoring System — Real-Time Sensor Control & Data Processing

ESP32 microcontroller, Gas sensor integration, Real-time data acquisition, Alert logic, Signal processing, Performance validation

- Designed real-time sensor control system with data acquisition, signal processing, and automated alert logic; implemented embedded control algorithms
- Debugged false alert triggering through systematic troubleshooting; applied signal filtering and validation logic demonstrating advanced control system problem-solving
- Conducted field testing and optimization demonstrating production-grade control system design (patent co-filed)

INDUSTRY EXPERIENCE

Picco2femto Semiconductor Devices — Assistant EDA Developer Intern (Feb 2026 - Present)

- Support semiconductor design verification and circuit analysis; identify and document design issues; contribute to design optimization processes

EDUCATION

SRM Institute of Science and Technology — B.Tech ECE · Batch 2027 · CGPA 9.16/10

CERTIFICATIONS & ACHIEVEMENTS

Published Research: ICIECST-2025 (Control systems for EV applications), JFinagles 2025 (Real-time IoT systems) | Patent co-filed on real-time monitoring system | Chip Design, AWS Fundamentals certifications | Technical Club Founder & Leader | Demonstrates technical excellence, control systems expertise, leadership, and continuous learning commitment